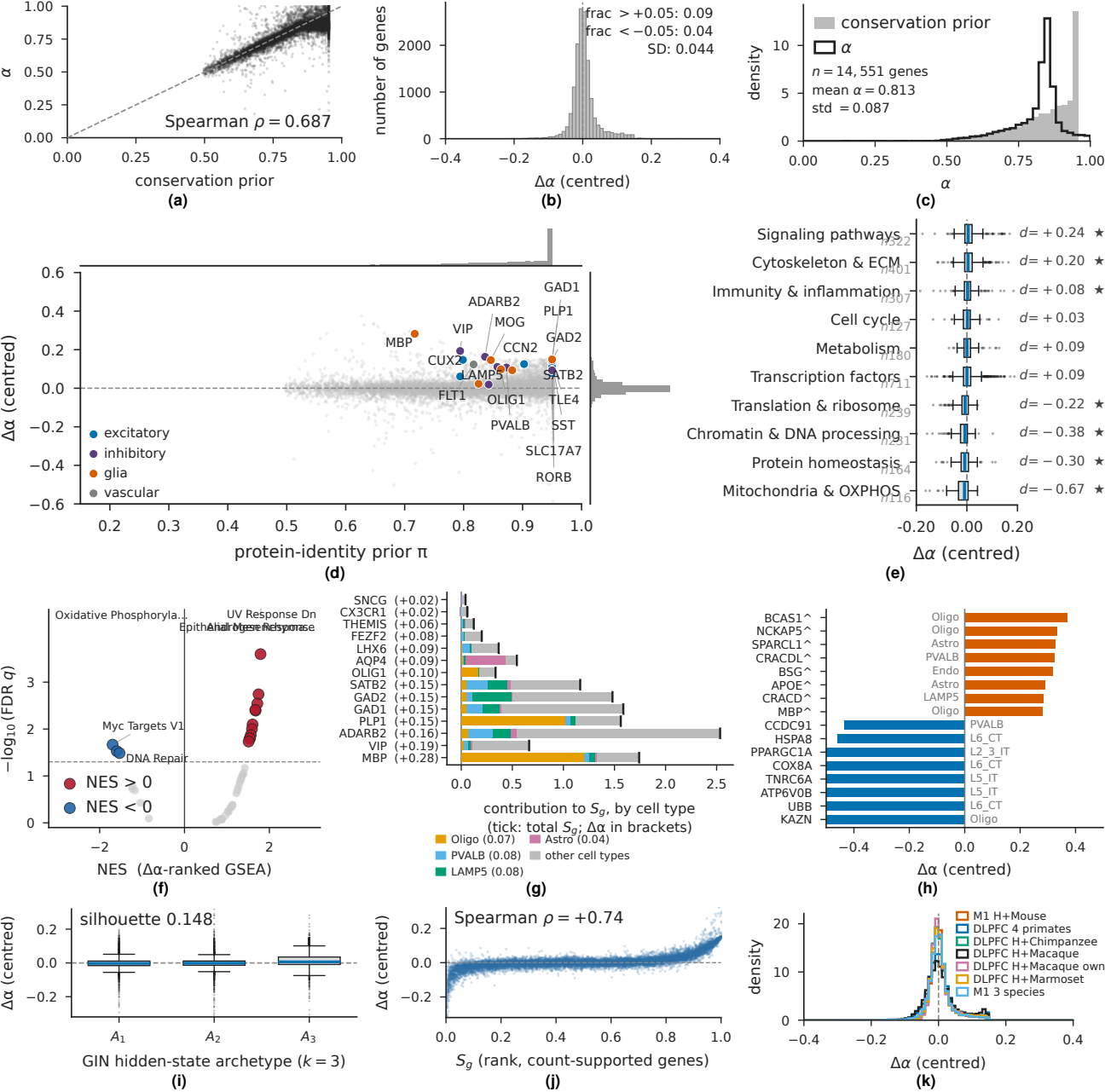

Spearman ρ = 0.687
conservation prior
α
(a)
frac > +0.05: 0.09
frac < −0.05: 0.04
SD: 0.044
number of genes
Δα (centred)
(b)
conservation prior
α
n = 14,551 genes
mean α = 0.813
std = 0.087
density
(c)
excitatory
inhibitory
glia
vascular
GAD1
ADARB2
PLP1
VIP
MOG
MBP
GAD2
CCN2
CUX2
SATB2
LAMP5
FLT1
OLIG1
TLE4
PVALB
SST
SLC17A7
RORB
Δα (centred)
protein-identity prior π
(d)
Signaling pathways n322 d = +0.24 ★
Cytoskeleton & ECM n401 d = +0.20 ★
Immunity & inflammation n307 d = +0.08 ★
Cell cycle n127 d = +0.03
Metabolism n180 d = +0.09
Transcription factors n711 d = +0.09
Translation & ribosome n239 d = −0.22 ★
Chromatin & DNA processing n251 d = −0.38 ★
Protein homeostasis n164 d = −0.30 ★
Mitochondria & OXPHOS n216 d = −0.67 ★
Δα (centred)
(e)
Oxidative Phosphoryla...
UV Response Dn
Myc Targets V1
DNA Repair
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
SNCG (+0.02)
CX3CR1 (+0.02)
THEMIS (+0.06)
FEZF2 (+0.08)
LHX6 (+0.09)
AQP4 (+0.09)
OLIG1 (+0.10)
SATB2 (+0.15)
GAD2 (+0.15)
GAD1 (+0.15)
PLP1 (+0.15)
ADARB2 (+0.16)
VIP (+0.19)
MBP (+0.28)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
Oligo (0.07)
Astro (0.04)
PVALB (0.08)
other cell types
LAMP5 (0.08)
(g)
BCAS1^ Oligo
NCKAP5^ Oligo
SPARCL1^ Astro
CRACDL^ PVALB
BSG^ Endo
APOE^ Astro
CRACD^ LAMP5
MBP^ Oligo
CCDC91 PVALB
HSPA8 L6_CT
PPARGC1A L2_3_IT
COX8A L6_CT
TNRC6A L5_IT
ATP6V0B L5_IT
UBB L6_CT
KAZN Oligo
Δα (centred)
(h)
silhouette 0.148
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.74
Δα (centred)
S_g (rank, count-supported genes)
(j)
M1 H+Mouse
DLPFC 4 primates
DLPFC H+Chimpanzee
DLPFC H+Macaque
DLPFC H+Macaque own
DLPFC H+Marmoset
M1 3 species
density
Δα (centred)
(k)